7	A driving force of 1.6 kN is produced by the car engine on the wheels to keep the car moving at a constant speed of 11 m s^{-1} on level road. The total energy of the fuel consumed by the engine is 1.8 MJ per minute. What is the efficiency of the car in converting the energy of the fuel into useful work done on the car to keep it moving at the constant speed of 11 m s^{-1} on level ground?							
	A	20 %	**B**	33 %	**C**	36 %	**D**	59 %
L2	**Answer: D** **Useful power output = $F_{driving}$ v = 1600 x 11 = 17600 W** **Power input of fuel = E / t = 1.8 x 10^6 / 60 = 30000 W** **Efficiency = (Useful power output / Power input of fuel) x 100 %** **= (17600 / 30000) x 100 % = 59 %** *Note: All of the power output goes into rate of work done against resistive forces on the car. No transformation into KE (since constant speed), and no transformation into GPE (since level ground).*							